Chapter 6. Estimation

Method of moments, maximum likelihood, multiparameter MLE, asymptotics of the MLE, least squares

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Setting

$X_1, \dots, X_n$ is a random sample from a pdf $f(x;\theta)$ with unknown parameter $\theta \in \Omega$. Guessing the value of θ, or of a function $\eta = \eta(\theta)$, from the sample is **estimation**. A statistic $\hat{\eta} = \hat{\eta}(X_1, \dots, X_n)$ used for this is an **estimator**, and its value at the data is an **estimate**. We write P_θ, E_θ, Var_θ to stress that probabilities are computed under the model with parameter θ. The most basic requirement is that $\hat{\eta}_n$ approach its target as n grows.

Definition (Consistency)

$\hat{\eta}_n$ is **consistent** for $\eta(\theta)$ if $\hat{\eta}_n \xrightarrow{p} \eta(\theta)$ under P_θ for every $\theta \in \Omega$, i.e. $\lim_n P_\theta(|\hat{\eta}_n - \eta(\theta)| \geq \epsilon) = 0$ for all $\epsilon > 0$ and all θ.

6.1 The method of moments

To estimate the population mean we use the sample mean; likewise the population moment $m_r = EX_1^r$ is naturally estimated by the sample moment $\hat{m}_r = \frac{1}{n}\sum X_i^r$. If a parameter can be written as $\eta = g(m_1, \dots, m_k)$, the **method-of-moments estimator** (MME) is

$$\hat{\eta}^{\mathrm{MME}} = g(\hat{m}_1, \dots, \hat{m}_k).$$

For a multivariate population one uses joint moments $m_{r,s} = E(X_1^r Y_1^s)$ and their sample versions $\hat{m}_{r,s} = \frac{1}{n}\sum X_i^r Y_i^s$ in the same way.

Example 6.1

(a) $\sigma^2 = m_2 - m_1^2$ and $\sigma = \sqrt{m_2 - m_1^2}$, so

$$\hat{\sigma}^{2\,\mathrm{MME}} = \hat{m}_2 - \hat{m}_1^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2, \qquad \hat{\sigma}^{\mathrm{MME}} = \sqrt{\frac{1}{n}\sum(X_i - \bar{X})^2}.$$

(b) The MME is not unique. For Poisson(λ), $\lambda = m_1 = m_2 - m_1^2$, which gives the two MMEs $\bar{X}$ and $\frac{1}{n}\sum(X_i - \bar{X})^2$. One normally uses the lowest moments possible.

(c) For a bivariate population, $\rho = \dfrac{m_{1,1} - m_{1,0}m_{0,1}}{\sqrt{m_{2,0} - m_{1,0}^2}\sqrt{m_{0,2} - m_{0,1}^2}}$. Substituting sample moments gives

$$\hat{\rho}^{\mathrm{MME}} = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum(X_i - \bar{X})^2}\sqrt{\sum(Y_i - \bar{Y})^2}},$$

the sample correlation coefficient.

(d) For Gamma(α, β), $m_1 = \alpha\beta$ and $m_2 - m_1^2 = \alpha\beta^2$. Solving, $\hat{\beta} = \hat{\sigma}^2/\bar{X}$ and $\hat{\alpha} = \bar{X}^2/\hat{\sigma}^2$ with $\hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$.

Theorem 6.1 (Consistency of MMEs)

If $E|X_1|^k < \infty$ and g is continuous at $(m_1, \dots, m_k)$, then $\hat{\eta}_n^{\mathrm{MME}} = g(\hat{m}_1, \dots, \hat{m}_k) \xrightarrow{p} g(m_1, \dots, m_k) = \eta$.

Proof. By the multivariate law of large numbers, $(\hat{m}_1, \dots, \hat{m}_k)' \xrightarrow{p} (m_1, \dots, m_k)'$ (Example 5.5 in Volume I). Continuous functions preserve convergence in probability (Theorem 5.7 in Volume I). ■

For example, $\hat{\sigma}^{2\,\mathrm{MME}}$ is consistent, and hence so is $S^2 = \frac{n}{n-1}\hat{\sigma}^{2\,\mathrm{MME}}$; the sample correlation is consistent for ρ. Consistency is the first property to check for any proposed estimator, and Theorem 6.1 shows that MMEs usually have it. They are also asymptotically normal:

Theorem 6.2 (Limiting distribution of MMEs)

If $EX_1^{2k} < \infty$ and g has continuous first partial derivatives at $\mathbf{m} = (m_1, \dots, m_k)'$, then

$$\sqrt{n}(\hat{\eta}_n^{\mathrm{MME}} - \eta) \xrightarrow{d} N\big(0,\ \dot{g}(\mathbf{m})'\Sigma\,\dot{g}(\mathbf{m})\big), \qquad \Sigma = (m_{r+s} - m_r m_s)_{r,s=1}^{k}.$$

Proof. The vectors $(X_i, X_i^2, \dots, X_i^k)'$ are i.i.d. with mean $\mathbf{m}$ and covariance matrix Σ, since $\mathrm{Cov}(X_1^r, X_1^s) = m_{r+s} - m_r m_s$. The multivariate CLT gives $\sqrt{n}(\hat{\mathbf{m}} - \mathbf{m}) \xrightarrow{d} N_k(\mathbf{0}, \Sigma)$, and the delta method (Theorem 5.12 in Volume I) finishes the proof. ∎

Example 6.2

For $\hat{\sigma}^{2\,\mathrm{MME}} = g(\hat{m}_1, \hat{m}_2)$ with $g(m_1, m_2) = m_2 - m_1^2$, $\dot{g} = (-2m_1, 1)'$, and a direct computation of $\dot{g}'\Sigma\dot{g}$ gives $\mu_4 - \sigma^4$, with $\mu_4 = E(X_1 - \mu)^4$. This agrees with Example 5.10 in Volume I.

6.2 Maximum likelihood

Suppose the model is $\{f(x;\theta) : \theta \in \Omega\}$. The probability that the sample falls in a small box around the observed $\mathbf{x} = (x_1, \dots, x_n)'$ is about $\prod_i f(x_i;\theta)\,|\Delta\mathbf{x}|$. Comparing $\prod f(x_i;\theta)$ with $\prod f(x_i;\theta')$ tells us which of two parameter values makes the observed data more plausible. So we regard the joint pdf as a function of θ with the data fixed.

Definition (Likelihood and MLE)

(a) The **likelihood function** is $L(\theta) = L(\theta;\mathbf{x}) = \prod_{i=1}^n f(x_i;\theta)$, $\theta \in \Omega$, and $l(\theta) = \log L(\theta) = \sum \log f(x_i;\theta)$ is the **log-likelihood**.

(b) A value $\hat{\theta}(\mathbf{x}) \in \Omega$ with $L(\hat{\theta}(\mathbf{x});\mathbf{x}) = \max_{\theta\in\Omega} L(\theta;\mathbf{x})$ is a **maximum likelihood estimate**; the statistic $\hat{\theta}^{\mathrm{MLE}} = \hat{\theta}(\mathbf{X})$ is the **maximum likelihood estimator** (MLE).

The derivative $\dot{l}(\theta)$ is easier to handle than that of L. The equation $\dot{l}(\theta) = 0$ is the **likelihood equation**, and the MLE is usually found among its roots.

Example 6.3 (Poisson)

For a Poisson(λ) sample, $\lambda \geq 0$ (with $\lambda = 0$ the point mass at 0),

$$l(\lambda) = -n\lambda + n\bar{x}\log\lambda - \log(x_1!\cdots x_n!), \qquad \dot{l}(\lambda) = -n + \frac{n\bar{x}}{\lambda}, \qquad \ddot{l}(\lambda) = -\frac{n\bar{x}}{\lambda^2}.$$

If $\bar{x} > 0$, l increases up to $\bar{x}$ and decreases afterwards, so $\max_{\lambda>0} l = l(\bar{x})$. If $\bar{x} = 0$, $l(\lambda) = -n\lambda$ is decreasing and the supremum is the limit at $\lambda = 0$. Since L is continuous on $[0,\infty)$, in both cases $\max_{\lambda\geq 0} L(\lambda) = L(\bar{x})$, and $\hat{\lambda}^{\mathrm{MLE}} = \bar{X}$.

Theorem 6.3

Let l be twice differentiable with continuous second derivative on an open interval Ω_o. If $\ddot{l}(\theta) < 0$ for all $\theta \in \Omega_o$ and $\dot{l}(\hat{\theta}) = 0$ for some $\hat{\theta} \in \Omega_o$, then $\hat{\theta}$ is the unique maximizer of l on

Ω_o.

Proof. By Taylor's theorem, for each $\theta \in \Omega_o$ there is θ^* between θ and $\hat\theta$ with

$$l(\theta) = l(\hat\theta) + \dot l(\hat\theta)(\theta-\hat\theta) + \tfrac12 \ddot l(\theta^*)(\theta-\hat\theta)^2 = l(\hat\theta) + \tfrac12 \ddot l(\theta^*)(\theta-\hat\theta)^2 < l(\hat\theta)$$

for $\theta \neq \hat\theta$. ■

Example 6.4 (Bernoulli)

For a Bernoulli(p) sample, $0 \le p \le 1$, with $\hat p = \bar x$,

$$l(p) = n\hat p \log p + n(1-\hat p)\log(1-p), \qquad \dot l(p) = \frac{n\hat p}{p} - \frac{n(1-\hat p)}{1-p}, \qquad \ddot l(p) = -\frac{n\hat p}{p^2} - \frac{n(1-\hat p)}{(1-p)^2}.$$

If $0 < \hat p < 1$, then $\ddot l < 0$ on $(0,1)$ and $\dot l(\hat p) = 0$, so $\hat p$ maximizes l by Theorem 6.3. If $\hat p = 0$ (all $x_i = 0$), $L(p) = (1-p)^n$ is maximized at $p = 0$; if $\hat p = 1$, $L(p) = p^n$ is maximized at $p = 1$. In all cases $\hat p^{\mathrm{MLE}} = \bar X$.

The condition $\ddot l < 0$ says that l is **strictly concave**: $l(\alpha\theta_1 + (1-\alpha)\theta_2) > \alpha l(\theta_1) + (1-\alpha) l(\theta_2)$ for $0 < \alpha < 1$ and $\theta_1 \neq \theta_2$. When the likelihood equation cannot be solved explicitly, the following guarantees a unique root that is the MLE.

Theorem 6.4

Let l be twice continuously differentiable on an open interval Ω_o, with $\ddot l(\theta) < 0$ on Ω_o and $l(\theta) \to -\infty$ as θ tends to either end of Ω_o. Then the likelihood equation $\dot l(\theta) = 0$ has exactly one root $\hat\theta$ in Ω_o, and $\hat\theta$ is the unique maximizer of l.

Proof. Consider $\Omega_o = (-\infty, \infty)$; other intervals are similar. Since $l \to -\infty$ at both ends and l is continuous, l attains its maximum at some $\hat\theta$. If it were attained at two points $\hat\theta_1 < \hat\theta_2$, strict concavity would give

$$l\big(\tfrac12\hat\theta_1 + \tfrac12\hat\theta_2\big) > \tfrac12 l(\hat\theta_1) + \tfrac12 l(\hat\theta_2) = \max l,$$

a contradiction. At the interior maximum $\dot l(\hat\theta) = 0$. Since $\dot l$ is strictly decreasing ($\ddot l < 0$), it has no other zero. ■

When the root has no closed form, it can be computed by **Newton–Raphson**: linearize $0 = \dot l(\hat\theta) \approx \dot l(\theta^{(r)}) + \ddot l(\theta^{(r)})(\hat\theta - \theta^{(r)})$ and iterate

$$\hat\theta^{(r+1)} = \hat\theta^{(r)} + \big[-\ddot l(\hat\theta^{(r)})\big]^{-1} \dot l(\hat\theta^{(r)}), \qquad r = 0, 1, \ldots,$$

starting, for instance, from the MME $\hat\theta^{(0)}$.

Example 6.5 (Logistic location)

For the logistic model $f(x;\theta) = e^{-(x-\theta)}/(1+e^{-(x-\theta)})^2$, $\theta \in \mathbb{R}$,

$$l(\theta) = -n\bar x + n\theta - 2\sum_{i=1}^{n} \log\big(1 + e^{-(x_i-\theta)}\big), \qquad \dot l(\theta) = n - 2\sum_i \frac{e^{-(x_i-\theta)}}{1+e^{-(x_i-\theta)}},$$

$$\ddot l(\theta) = -2\sum_i \frac{e^{-(x_i-\theta)}}{(1+e^{-(x_i-\theta)})^2} < 0.$$

As $\theta \to -\infty$, $l(\theta) \approx n\theta - n\bar x \to -\infty$; as $\theta \to +\infty$, $l(\theta) \approx n\theta - 2\sum(\theta - x_i) - n\bar x = -n\theta + n\bar x \to$

$-\infty$. By Theorem 6.4 the MLE is the unique root of $\dot{l}(\theta) = 0$. The logistic mean is θ, so Newton–Raphson can start at $\hat{\theta}^{(0)} = \bar{x}$.

Theorem 6.5 (Invariance)

If $\hat{\theta}^{\mathrm{MLE}}$ exists and $\eta = g(\theta)$ is a one-to-one function on Ω, then $\hat{\eta}^{\mathrm{MLE}} = g(\hat{\theta}^{\mathrm{MLE}})$.

Proof. Reparametrizing does not change the family of densities: the density with parameter $\eta \in g(\Omega)$ is $f(\cdot; g^{-1}(\eta))$. So the likelihood in η is $L^*(\eta) = L(g^{-1}(\eta))$, and

$$\max_{\eta \in g(\Omega)} L^*(\eta) = \max_{\theta \in \Omega} L(\theta) = L(\hat{\theta}^{\mathrm{MLE}}) = L^*(g(\hat{\theta}^{\mathrm{MLE}})).$$

■

Example 6.6 (Exponential)

For $\mathrm{Exp}(\theta)$, $f(x;\theta) = \theta^{-1}e^{-x/\theta}$, we have $l(\theta) = -n\log\theta - n\bar{x}/\theta$ and $\ddot{l}(\theta) = n/\theta^2 - 2n\bar{x}/\theta^3$, which is not negative everywhere; Theorem 6.4 does not apply directly. With $\lambda = 1/\theta$, however, $l(\lambda) = n\log\lambda - n\bar{x}\lambda$ has $\ddot{l}(\lambda) = -n/\lambda^2 < 0$, and $l \to -\infty$ as $\lambda \to 0$ or ∞ (when $\bar{x} > 0$). So $\hat{\lambda} = 1/\bar{X}$, and by invariance $\hat{\theta}^{\mathrm{MLE}} = 1/\hat{\lambda} = \bar{X}$.

A good parametrization can make the log-likelihood concave. For exponential families there is a canonical choice.

Theorem 6.6 (MLE in a one-parameter exponential family)

Let

$$f(x;\eta) = \exp\{\eta T(x) - A(\eta) + S(x)\}, \qquad x \in \mathcal{X},\ \eta \in N,$$

where (i) the support $\mathcal{X}$ does not depend on η;
(ii) N is an open interval;
(iii) $\mathrm{Var}_\eta\, T(X_1) > 0$. If the likelihood equation

$$\dot{A}(\eta) = \frac{1}{n}\sum_{i=1}^{n} T(x_i), \qquad \text{i.e.} \quad E_\eta T(X_1) = \frac{1}{n}\sum_{i=1}^{n} T(x_i),$$

has a root $\hat{\eta}$ in N, then $\hat{\eta}$ is the MLE of η.

Proof. The key fact is the cgf of $T(X_1)$. In the continuous case, for $|s|$ small enough that $s+\eta \in N$ (possible by (ii)),

$$E_\eta e^{sT(X_1)} = \int e^{(s+\eta)T(x)-A(\eta)+S(x)}dx = e^{A(s+\eta)-A(\eta)}\int f(x; s+\eta)\,dx = e^{A(s+\eta)-A(\eta)},$$

using (i) for the last integral. So the cgf is $A(s+\eta) - A(\eta)$, and differentiating at $s = 0$ gives $E_\eta T(X_1) = \dot{A}(\eta)$ and $\mathrm{Var}_\eta\, T(X_1) = \ddot{A}(\eta)$, which is positive by (iii). The log-likelihood $l(\eta) = \eta\sum T(x_i) - nA(\eta) + \sum S(x_i)$ therefore has $\ddot{l}(\eta) = -n\ddot{A}(\eta) < 0$, and a root of $\dot{l}(\eta) = \sum T(x_i) - n\dot{A}(\eta) = 0$ is the MLE by Theorem 6.3. ■

If the family is written in another parameter, $f(x;\theta) = \exp\{g(\theta)T(x) - A(g(\theta)) + S(x)\}$ with $\eta = g(\theta)$ one-to-one, then by invariance the MLE of θ solves $E_\theta T(X_1) = \frac{1}{n}\sum T(x_i)$: *match the mean of T to its sample average.*

Example 6.7

(a) Bernoulli(p): $f = \exp\{x \log \frac{p}{1-p} + \log(1-p)\}$, $T(x) = x$. The equation $E_p X = p = \bar{x}$ gives $\hat{p} = \bar{x}$ when $0 < \bar{x} < 1$.

(b) Poisson(λ): $T(x) = x$, $\eta = \log \lambda$; $\hat{\lambda} = \bar{x}$ when $\bar{x} > 0$.

(c) Geometric, $f(x;p) = (1-p)^{x-1}p$, $x = 1, 2, \ldots$: $T(x) = x$ with $\eta = \log(1-p)$. Then $E_p X = 1/p = \bar{x}$ gives $\hat{p} = 1/\bar{x}$ when $\bar{x} > 1$.

(d) Exponential(θ): $T(x) = x$, $\eta = -1/\theta$; $\hat{\theta} = \bar{x}$.

(e) Pareto, $f(x;\theta) = \theta x^{-\theta-1}$, $x > 1$: $f = \exp\{-\theta \log x + \log \theta - \log x\}$, so $T(x) = -\log x$. Since $X^{-\theta} \sim U(0,1)$, $\theta \log X \sim \text{Exp}(1)$ and $E_\theta(-\log X) = -1/\theta$. The equation gives $\hat{\theta} = 1/(\frac{1}{n}\sum \log x_i)$.

In the examples so far the likelihood was differentiable. Otherwise one studies the likelihood directly, and the MLE need not be unique.

Example 6.8 (Double exponential)

For $f(x;\theta) = \frac{1}{2}e^{-|x-\theta|}$ and $n = 2m+1$, $l(\theta) = -\sum|x_i - \theta| - n \log 2$. With order statistics $x_{(1)} < \cdots < x_{(n)}$ and $x_{(0)} = -\infty$, $x_{(n+1)} = \infty$, for $x_{(r)} \le \theta < x_{(r+1)}$,

$$l(\theta) = (n-2r)\theta + \sum_{i \le r} x_{(i)} - \sum_{i > r} x_{(i)} - n \log 2.$$

The slope $n - 2r$ is positive for $r \le m$ and negative for $r \ge m+1$, so l increases up to $x_{(m+1)}$ and decreases afterwards. The MLE is the sample median $X_{(m+1)}$.

Example 6.9 (Uniform)

(a) For $U(0,\theta)$, $L(\theta) = \theta^{-n}\mathbf{1}\{\theta \ge x_{(n)}\}$, which is decreasing on $[x_{(n)}, \infty)$ and zero before it. So $\hat{\theta}^{\text{MLE}} = X_{(n)}$; the likelihood equation plays no role.
(b) For $U(\theta-1, \theta+1)$, $L(\theta) = 2^{-n}\mathbf{1}\{x_{(n)} - 1 \le \theta \le x_{(1)} + 1\}$. Every point of $[x_{(n)} - 1, x_{(1)} + 1]$ maximizes L, so every statistic between $X_{(n)} - 1$ and $X_{(1)} + 1$, for instance $\alpha(X_{(n)} - 1) + (1 - \alpha)(X_{(1)} + 1)$ with $0 \le \alpha \le 1$, is an MLE. Other criteria (efficiency, simplicity) must choose among them.

6.3 MLE for vector parameters

For $\theta = (\theta_1, \ldots, \theta_k)' \in \Omega \subseteq \mathbb{R}^k$ the definitions are the same: $L(\theta) = \prod f(x_i;\theta)$, and $\hat{\theta}^{\text{MLE}}$ maximizes L over Ω. Its components are the MLEs of the components of θ. One can maximize one variable at a time, or solve the likelihood equations $\partial l / \partial \theta_j = 0$.

Example 6.10 (Double exponential with scale)

For $f(x;\mu,\sigma) = \frac{1}{2\sigma}e^{-|x-\mu|/\sigma}$ and $n = 2m+1$,

$$l(\mu,\sigma) = -\frac{1}{\sigma}\sum|x_i - \mu| - n \log \sigma - n \log 2.$$

For each σ, l is maximized in μ by the median $\hat{\mu} = x_{(m+1)}$ (Example 6.8). Then

$$\frac{\partial}{\partial \sigma} l(\hat{\mu}, \sigma) = \frac{\sum|x_i - \hat{\mu}|}{\sigma^2} - \frac{n}{\sigma} > 0 \iff \sigma < \frac{1}{n}\sum|x_i - \hat{\mu}|,$$

so $\hat{\sigma} = \frac{1}{n}\sum|x_i - \hat{\mu}|$. Since $\max_\sigma \max_\mu l = \max_\sigma l(\hat{\mu}, \sigma)$, the MLE is $(X_{(m+1)}, \frac{1}{n}\sum|X_i - X_{(m+1)}|)$.

Example 6.11 (Normal)

For $N(\mu, \sigma^2)$,

$$l(\mu, \sigma^2) = -\frac{1}{2\sigma^2}\sum(x_i - \mu)^2 - \frac{n}{2}\log\sigma^2 - \frac{n}{2}\log 2\pi.$$

Since $\sum(x_i - \mu)^2 = \sum(x_i - \bar{x})^2 + n(\bar{x} - \mu)^2$, l is maximized in μ at $\bar{x}$ for every σ^2. Then $\theta_2 = \sigma^2 \mapsto l(\bar{x}, \theta_2)$ has derivative $\frac{1}{2\theta_2^2}\sum(x_i - \bar{x})^2 - \frac{n}{2\theta_2}$, positive iff $\theta_2 < \frac{1}{n}\sum(x_i - \bar{x})^2$. Hence $\hat{\mu} = \bar{X}$ and $\hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$.

Theorem 6.7 (Invariance for functions)

If $\hat{\theta}^{\text{MLE}}$ exists and $\eta_1 = g_1(\theta)$ can be completed to a one-to-one map $\eta = (g_1(\theta), g_2(\theta))$, then $\hat{\eta}_1^{\text{MLE}} = g_1(\hat{\theta}^{\text{MLE}})$.

Proof. By Theorem 6.5 (whose proof works verbatim for vectors), $\hat{\eta}^{\text{MLE}} = (g_1(\hat{\theta}^{\text{MLE}}), g_2(\hat{\theta}^{\text{MLE}}))$, and the MLE of a component is the corresponding component of the MLE. ■

Example 6.12

(a) In $N(\mu, \sigma^2)$, $(|\mu|, \operatorname{sgn}\mu, \sigma^2)$ is a one-to-one function of (μ, σ^2), so $\widehat{|\mu|}^{\text{MLE}} = |\bar{X}|$.
(b) In Bernoulli(p), $(p(1-p), \operatorname{sgn}(p - \frac{1}{2}))$ is one-to-one in p, so the MLE of the variance $p(1-p)$ is $\hat{p}(1-\hat{p})$ with $\hat{p} = \bar{X}$.

The one-dimensional results on concavity extend; their proofs are the same and are omitted.

Theorem 6.8

Let $\Omega_o = (a_1, b_1) \times \cdots \times (a_k, b_k)$ and let l have continuous second partial derivatives, with Hessian $\ddot{l}(\theta) = (\partial^2 l / \partial\theta_i \partial\theta_j)$.

(a) If $\mathbf{c}'\ddot{l}(\theta)\mathbf{c} < 0$ for all $\mathbf{c} \neq \mathbf{0}$ and $\theta \in \Omega_o$, and $\dot{l}(\hat{\theta}) = \mathbf{0}$ for some $\hat{\theta} \in \Omega_o$, then $\hat{\theta}$ is the unique maximizer of l.

(b) If, moreover, $l(\theta) \to -\infty$ as θ tends to the boundary of Ω_o, then $\dot{l}(\theta) = \mathbf{0}$ has exactly one root in Ω_o, and it is the MLE.

Theorem 6.9 (Multiparameter exponential family)

Let $f(x; \eta) = \exp\{\sum_{j=1}^k \eta_j T_j(x) - A(\eta) + S(x)\}$, $x \in \mathcal{X}$, $\eta \in N$, where (i) $\mathcal{X}$ does not depend on η;
(ii) N contains an open box in $\mathbb{R}^k$;
(iii) $\operatorname{Var}_\eta(\mathbf{c}'\mathbf{T}(X_1)) > 0$ for every $\mathbf{c} \neq \mathbf{0}$. If the likelihood equations

$$E_\eta T_j(X_1) = \frac{1}{n}\sum_{i=1}^n T_j(x_i), \qquad j = 1, \ldots, k,$$

have a root $\hat{\eta}$ in N, then $\hat{\eta}$ is the MLE.

Proof. As in Theorem 6.6, $\mathbf{T}(X_1)$ has cgf $A(\mathbf{s}+\eta)-A(\eta)$, so $E_\eta \mathbf{T} = \dot{A}(\eta)$ and $\operatorname{Var}_\eta \mathbf{T} = \ddot{A}(\eta)$. Then $\mathbf{c}'\ddot{l}(\eta)\mathbf{c} = -n\mathbf{c}'\ddot{A}(\eta)\mathbf{c} = -n\operatorname{Var}_\eta(\mathbf{c}'\mathbf{T}) < 0$, and Theorem 6.8(a) applies. ∎

Example 6.13

(a) *Normal.* $N(\mu,\sigma^2)$ has $\eta_1 = \mu/\sigma^2$, $\eta_2 = -1/(2\sigma^2)$, $T = (x, x^2)$. The equations $EX = \bar{x}$, $EX^2 = \frac{1}{n}\sum x_i^2$ give $\hat{\mu} = \bar{x}$ and $\hat{\sigma}^2 + \hat{\mu}^2 = \frac{1}{n}\sum x_i^2$, the same answer as Example 6.11.
(b) *Bivariate normal.* The exponent is linear in $T = (x, y, xy, x^2, y^2)$. Matching the five means gives $\hat{\mu}_1 = \bar{x}$, $\hat{\mu}_2 = \bar{y}$, $\hat{\sigma}_j^2 = \frac{1}{n}\sum(\cdot - \text{mean})^2$, and $\hat{\rho}^{\text{MLE}}$ equal to the sample correlation coefficient.
(c) *Multinomial.* For $\text{Multi}(n, \mathbf{p})$ observed once, drop the redundant last cell: with $\eta_j = \log(p_j/p_k)$, $j < k$, the pmf is $\exp\{\sum_{j<k}\eta_j x_j - nA(\eta) + S(x)\}$ with $A(\eta) = \log(1 + e^{\eta_1} + \cdots + e^{\eta_{k-1}})$. Matching $EX_j = np_j$ to x_j gives $\hat{p}_j = x_j/n$ (when all $0 < x_j < n$).

6.4 Asymptotic properties of the MLE

Example 6.14

In the examples above the MLE coincided with an MME, so the results of Chapter 5 apply:

- Bernoulli: $\hat{p} \xrightarrow{p} p$ and $\sqrt{n}(\hat{p}-p) \xrightarrow{d} N(0, p(1-p))$.
- Poisson: $\sqrt{n}(\hat{\lambda}-\lambda) \xrightarrow{d} N(0,\lambda)$. Exponential: $\sqrt{n}(\hat{\theta}-\theta) \xrightarrow{d} N(0,\theta^2)$.
- Normal: $\sqrt{n}(\hat{\mu}-\mu) \sim N(0,\sigma^2)$ exactly, $n\hat{\sigma}^2/\sigma^2 \sim \chi^2(n-1)$, and $\sqrt{n}(\hat{\sigma}^2-\sigma^2) \xrightarrow{d} N(0, 2\sigma^4)$.
- Bivariate normal: $\sqrt{n}(\hat{\rho}-\rho) \xrightarrow{d} N(0, (1-\rho^2)^2)$.

In general the MLE is not an MME, and we need a direct argument. Write $\bar{l}_n(\theta) = \frac{1}{n}\sum_i \log f(X_i;\theta)$ for the average log-likelihood and let θ^0 be the true value. By the law of large numbers $\bar{l}_n(\theta) \xrightarrow{p} E_{\theta^0}\log f(X_1;\theta)$ for each θ. So we expect $\hat{\theta}_n = \arg\max \bar{l}_n$ to approach $\arg\max_\theta E_{\theta^0}\log f(X_1;\theta)$. The next theorem shows that this maximizer is θ^0.

Theorem 6.10 (Kullback–Leibler divergence)

Assume (R0) *identifiability*: $f(\cdot;\theta) = f(\cdot;\theta^0)$ implies $\theta = \theta^0$; and (R1) the support $\{x : f(x;\theta) > 0\} = \mathcal{X}$ does not depend on θ. Let

$$KL(\theta,\theta^0) = -E_{\theta^0}\log\frac{f(X;\theta)}{f(X;\theta^0)}.$$

Then (i) $KL(\theta,\theta^0) \geq 0$ for all θ, and (ii) $KL(\theta,\theta^0) = 0$ iff $\theta = \theta^0$. Equivalently, θ^0 is the unique maximizer of $E_{\theta^0}\log f(X;\theta)$.

Proof. For $t > 0$, $-\log t \geq 1 - t$, with equality iff $t = 1$. Put $t = f(X;\theta)/f(X;\theta^0)$ and take E_{θ^0}:

$$KL(\theta,\theta^0) \geq E_{\theta^0}\Big[1 - \frac{f(X;\theta)}{f(X;\theta^0)}\Big] = 1 - \int_{\mathcal{X}} f(x;\theta)\,dx = 0,$$

using (R1). Equality holds iff $f(X;\theta) = f(X;\theta^0)$ with probability one, i.e. (up to null sets) $f(\cdot;\theta) = f(\cdot;\theta^0)$, which by (R0) means $\theta = \theta^0$. ∎

To turn the heuristic into a proof one needs more than pointwise convergence: e.g. uniform convergence of $\bar{l}_n$ on compact sets and continuity of $E_{\theta^0} \log f(X_1;\theta)$. When the log-likelihood is strictly concave and tends to $-\infty$ at the boundary, these hold, and one obtains:

Theorem 6.11 (Consistency of the MLE)

Let Ω be an open box in $\mathbb{R}^k$ and assume (R0), (R1). Suppose l_n has continuous second partial derivatives with $\mathbf{c}'\ddot{l}_n(\theta)\mathbf{c} < 0$ for $\mathbf{c} \neq \mathbf{0}$ and $l_n(\theta) \to -\infty$ at the boundary of Ω, and that $E_{\theta^0} \log f(X_1;\theta)$ exists and is continuous in θ. Then the MLE $\hat{\theta}_n$ is the unique root of $\dot{l}_n(\theta) = \mathbf{0}$, and $\hat{\theta}_n \xrightarrow{p} \theta$ under P_θ for every $\theta \in \Omega$.

(The proof is beyond these notes.) For example, the logistic MLE of Example 6.5 and the gamma MLE (the root of $-\Psi(\alpha) - \log\beta + \overline{\log x} = 0$, $-\alpha/\beta + \bar{x}/\beta^2 = 0$ with $\Psi = (\log\Gamma)'$) are consistent.

Heuristics for the limit law. Expanding the likelihood equation around θ,

$$0 = \dot{l}_n(\hat{\theta}_n) \approx \dot{l}_n(\theta) + \ddot{l}_n(\theta)(\hat{\theta}_n - \theta), \qquad \text{so} \qquad \sqrt{n}(\hat{\theta}_n - \theta) \approx \{-\ddot{l}_n(\theta)\}^{-1}\sqrt{n}\,\dot{l}_n(\theta).$$

By the LLN, $-\ddot{l}_n(\theta) \xrightarrow{p} I(\theta) := E_\theta[-\partial_\theta^2 \log f(X_1;\theta)]$, and by the CLT,

$$\sqrt{n}\,\dot{l}_n(\theta) \xrightarrow{d} N\big(0, \operatorname{Var}_\theta \partial_\theta \log f(X_1;\theta)\big),$$

provided $E_\theta \partial_\theta \log f(X_1;\theta) = 0$. We now make this rigorous.

Regularity conditions. In addition to (R0) and (R1):

(R2) Ω is an open subset of $\mathbb{R}^k$.

(R3) For every $x \in \mathcal{X}$, $\log f(x;\theta)$ has continuous first and second partial derivatives in θ.

(R4) For statistics $u(\mathbf{X})$ with $E_\theta u(\mathbf{X})$ finite, derivatives in θ up to second order may be taken under the integral (or sum) sign: $\partial_\theta^r E_\theta u(\mathbf{X}) = \int u(\mathbf{x}) \partial_\theta^r \prod f(x_i;\theta)\, d\mathbf{x}$, $r = 1, 2$.

(R5) The **Fisher information** $I(\theta) = \operatorname{Var}_\theta\left(\partial_\theta \log f(X_1;\theta)\right)$ is finite and nonsingular.

The derivative $\dot{l}_n(\theta)$ is called the **score function**.

Theorem 6.12 (Properties of the score)

Under (R0)–(R5),

(a) $E_\theta\{\partial_\theta \log f(X_1;\theta)\} = 0$;

(b) $I(\theta) = \operatorname{Var}_\theta\{\partial_\theta \log f(X_1;\theta)\} = E_\theta\{-\partial_\theta^2 \log f(X_1;\theta)\}$.

Proof. We treat a continuous one-dimensional θ. For all θ, $\int_{\mathcal{X}} \exp\{\log f(x;\theta)\}\, dx = 1$ (R1). Differentiating under the integral (R3, R4),

$$\int \{\partial_\theta \log f(x;\theta)\} f(x;\theta)\, dx = 0, \qquad \text{i.e.} \quad E_\theta \partial_\theta \log f(X_1;\theta) = 0.$$

Differentiating again,

$$\int \left\{\partial_\theta^2 \log f(x;\theta) + \left[\partial_\theta \log f(x;\theta)\right]^2\right\} f(x;\theta)\, dx = 0,$$

so $E_\theta[(\partial_\theta \log f)^2] = -E_\theta[\partial_\theta^2 \log f]$. By (a), the left side is the variance. ∎

Theorem 6.13 (Asymptotic normality of the MLE)

Assume (R0)–(R5) and also:

(R6) the MLE $\hat\theta_n$ is the unique root of $\dot l_n(\theta)=0$ and is consistent;

(R7) the third derivatives $\partial_\theta^3 \log f(x;\theta)$ exist and are continuous, and there is $M(x)$ with $\max_{\theta\in\Omega}\|\partial_\theta^3 \log f(x;\theta)\| \le M(x)$ and $E_\theta M(X_1)<\infty$.

Then $\sqrt{n}(\hat\theta_n-\theta)\xrightarrow{d} N(0, I(\theta)^{-1})$.

Proof. We give the one-dimensional proof. Write $\dot l_n = \dot l_n/n$, $\ddot l_n = \ddot l_n/n$ and $\bar l_n^{(3)} = l_n^{(3)}/n$.

Step 1 (expansion). By Taylor's theorem there is θ_n^* between θ and $\hat\theta_n$ with

$$0=\dot{\bar l}_n(\hat\theta_n)=\dot{\bar l}_n(\theta)+\ddot{\bar l}_n(\theta)(\hat\theta_n-\theta)+\tfrac12 \bar l_n^{(3)}(\theta_n^*)(\hat\theta_n-\theta)^2.$$

Step 2 (the remainder is negligible). Let $R_n=\frac12\bar l_n^{(3)}(\theta_n^*)(\hat\theta_n-\theta)$. For $\epsilon, K>0$,

$$P_\theta(|R_n|\ge\epsilon)\le P_\theta(|\bar l_n^{(3)}(\theta_n^*)|>K)+P_\theta(|\hat\theta_n-\theta|\ge 2\epsilon/K).$$

By (R7) and Markov's inequality, $P_\theta(|\bar l_n^{(3)}(\theta_n^*)|>K)\le P_\theta(\frac1n\sum M(X_i)>K)\le E_\theta M(X_1)/K$. By (R6), the second term tends to 0. So $\limsup_n P_\theta(|R_n|\ge\epsilon)\le E_\theta M(X_1)/K$ for every K, and $R_n\xrightarrow{p}0$.

Step 3 (linearization). Step 1 reads $0=\dot{\bar l}_n(\theta)+\{\ddot{\bar l}_n(\theta)+R_n\}(\hat\theta_n-\theta)$. By the LLN and Theorem 6.12(b), $-\ddot{\bar l}_n(\theta)\xrightarrow{p} I(\theta)$. Hence

$$\sqrt{n}(\hat\theta_n-\theta)=\big(I(\theta)+r_n\big)^{-1}\sqrt{n}\,\dot{\bar l}_n(\theta),\qquad r_n\xrightarrow{p}0.$$

Step 4 (limit). $\sqrt{n}\,\dot{\bar l}_n(\theta)=\sqrt{n}(\frac1n\sum\partial_\theta\log f(X_i;\theta)-0)\xrightarrow{d}N(0,I(\theta))$ by the CLT and Theorem 6.12. By Slutsky's theorem, $\sqrt{n}(\hat\theta_n-\theta)\xrightarrow{d} I(\theta)^{-1}Z$ with $Z\sim N(0,I(\theta))$, i.e. $N(0,I(\theta)^{-1})$. ∎

Condition (R7) is stronger than necessary but keeps the proof short. Step 3 also shows

$$\sqrt{n}(\hat\theta_n-\theta)=I(\theta)^{-1}\sqrt{n}\,\dot{\bar l}_n(\theta)+o_p(1):$$

the error of the MLE is driven by the score, and its precision grows with the information $I(\theta)$.

When the MLE has no closed form, one Newton–Raphson step from a good starting value is already as good asymptotically.

Theorem 6.14 (One-step estimator)

Assume (R0)–(R7), and let $\hat\theta_n^{(0)}$ be an estimator with $\sqrt{n}(\hat\theta_n^{(0)}-\theta)\xrightarrow{d}Z$ for some random variable Z (R8). Then

$$\hat\theta_n^{(1)}=\hat\theta_n^{(0)}+\big[-\ddot l_n(\hat\theta_n^{(0)})\big]^{-1}\dot l_n(\hat\theta_n^{(0)})$$

satisfies $\sqrt{n}(\hat\theta_n^{(1)}-\theta)\xrightarrow{d}N(0,I(\theta)^{-1})$.

The proof follows the same steps as Theorem 6.13, expanding $\dot l_n(\hat\theta_n^{(0)})$ around θ.

Example 6.15 (Logistic location)

In Example 6.5, start from $\hat\theta^{(0)} = \bar X$, which satisfies $\sqrt{n}(\bar X - \theta) \xrightarrow{d} N(0, \pi^2/3)$ (the logistic variance). The information is

$$I(\theta) = E_\theta\big[-\partial_\theta^2 \log f(X;\theta)\big] = 2E\Big[\frac{e^{-(X-\theta)}}{(1+e^{-(X-\theta)})^2}\Big] = 2E[U(1-U)] = \frac{1}{3},$$

since $U = 1/(1+e^{-(X-\theta)}) \sim U(0,1)$ by the probability integral transform. So the one-step estimator has $\sqrt{n}(\hat\theta^{(1)} - \theta) \xrightarrow{d} N(0,3)$, an improvement over $\pi^2/3 \approx 3.29$ for $\bar X$.

Example 6.16 (Gamma)

For Gamma(α,β), with $\Psi = (\log\Gamma)'$,

$$\dot l_n = \begin{pmatrix} -n\Psi(\alpha) - n\log\beta + n\overline{\log x} \\ -n\alpha/\beta + n\bar x/\beta^2 \end{pmatrix}, \qquad I(\theta) = \begin{pmatrix} \dot\Psi(\alpha) & 1/\beta \\ 1/\beta & \alpha/\beta^2 \end{pmatrix}.$$

Start from the MME $(\hat\alpha^{(0)}, \hat\beta^{(0)})$ of Example 6.1(d), which is $\sqrt{n}$-consistent by Theorem 6.2. One Newton step gives an estimator with limit $N_2(\mathbf{0}, I(\theta)^{-1})$, where

$$I(\theta)^{-1} = \frac{1}{\alpha\dot\Psi(\alpha) - 1}\begin{pmatrix} \alpha & -\beta \\ -\beta & \beta^2\dot\Psi(\alpha) \end{pmatrix}.$$

6.5 Least squares

In the **linear regression model**

$$Y_i = x_{i0}\beta_0 + x_{i1}\beta_1 + \cdots + x_{ip}\beta_p + e_i, \qquad Ee_i = 0,\ \operatorname{Var} e_i = \sigma^2,\ \operatorname{Cov}(e_i, e_j) = 0\ (i \neq j),$$

the mean response is a linear function of the explanatory variables. The **least squares estimator** (LSE) $\hat\beta$ minimizes $\sum_i\{Y_i - (x_{i0}\beta_0 + \cdots + x_{ip}\beta_p)\}^2$, and the mean response at $(x_0,\ldots,x_p)$ is estimated by $x_0\hat\beta_0 + \cdots + x_p\hat\beta_p$. In matrix form, $\mathbf{Y} = X\beta + \mathbf{e}$ with X an $n \times (p+1)$ matrix of rank $p+1$, $E\mathbf{e} = \mathbf{0}$, $\operatorname{Var}\mathbf{e} = \sigma^2 I_n$, and

$$\hat\beta^{\mathrm{LSE}} = \operatorname*{arg\,min}_{\beta\in\mathbb{R}^{p+1}} \|\mathbf{Y} - X\beta\|^2.$$

Theorem 6.15 (Least squares)

(a) $\Pi = X(X'X)^{-1}X'$ satisfies $\Pi' = \Pi$, $\Pi'(I-\Pi) = 0$ and $\Pi X = X$.

(b) $\|\mathbf{Y} - X\beta\|^2 = \|\Pi(\mathbf{Y} - X\beta)\|^2 + \|(I-\Pi)\mathbf{Y}\|^2$; hence $X\hat\beta^{\mathrm{LSE}} = \Pi\mathbf{Y}$ and $\hat\beta^{\mathrm{LSE}} = (X'X)^{-1}X'\mathbf{Y}$.

Proof. (a) Since $X'X$ is symmetric and invertible, $\Pi' = X\{(X'X)^{-1}\}'X' = \Pi$ and

$$\Pi'\Pi = X(X'X)^{-1}X'X(X'X)^{-1}X' = \Pi,$$

so $\Pi'(I-\Pi) = 0$, and $\Pi X = X$.

(b) By (a), $\mathbf{Y} - X\beta = \Pi(\mathbf{Y} - X\beta) + (I-\Pi)\mathbf{Y}$, and the cross term $2\{\Pi(\mathbf{Y} - X\beta)\}'(I-\Pi)\mathbf{Y}$ vanishes. The second term does not involve β, and the first is zero iff $X\beta = \Pi\mathbf{Y}$, i.e. $\beta = (X'X)^{-1}X'\mathbf{Y}$ (X has full column rank). ∎

Π is the orthogonal projection onto the column space of X. The error variance is estimated by the mean squared residual

$$\hat\sigma^2 = \frac{\|\mathbf{Y} - X\hat\beta\|^2}{n-p-1}.$$

Theorem 6.16 (Properties of the LSE)

(a) $E\hat\beta = \beta$ and $\operatorname{Var}\hat\beta = \sigma^2(X'X)^{-1}$.

(b) $E\hat\sigma^2 = \sigma^2$.

(c) If $\mathbf{e} \sim N_n(\mathbf{0}, \sigma^2 I)$, then $\hat\beta \sim N(\beta, \sigma^2(X'X)^{-1})$, $(n-p-1)\hat\sigma^2/\sigma^2 \sim \chi^2(n-p-1)$, and they are independent.

Proof. (a) $E\hat\beta = (X'X)^{-1}X'X\beta = \beta$ and $\operatorname{Var}\hat\beta = (X'X)^{-1}X'(\sigma^2 I)X(X'X)^{-1} = \sigma^2(X'X)^{-1}$.

(b) $(I-\Pi)\mathbf{Y} = (I-\Pi)\mathbf{e}$ since $(I-\Pi)X = 0$, so $(n-p-1)\hat\sigma^2 = \mathbf{e}'(I-\Pi)\mathbf{e}$. Using $\operatorname{tr}(AB) = \operatorname{tr}(BA)$,

$$E\,\mathbf{e}'(I-\Pi)\mathbf{e} = E\operatorname{tr}\{(I-\Pi)\mathbf{e}\mathbf{e}'\} = \operatorname{tr}\{(I-\Pi)\sigma^2 I\} = \sigma^2\{n - \operatorname{tr}((X'X)^{-1}X'X)\} = \sigma^2(n-p-1).$$

(c) is Theorem 4.19 in Volume I. ∎

Orthogonalization. Computation and interpretation are easiest when the columns of X are orthogonal. Partition $X = (X_0, X_1)$ with X_0 of size $n \times p_0$ and X_1 of size $n \times p_1$, and let $\Pi_0 = X_0(X_0'X_0)^{-1}X_0'$. Then $X_{1|0} = (I-\Pi_0)X_1$ is orthogonal to the columns of X_0, and

$$X\beta = X_0\beta_0 + X_1\beta_1 = X_0\gamma_0 + X_{1|0}\beta_1, \qquad \gamma_0 = \beta_0 + (X_0'X_0)^{-1}X_0'X_1\beta_1.$$

With $\Pi_{1|0} = X_{1|0}(X_{1|0}'X_{1|0})^{-1}X_{1|0}'$, the projection onto the column space of X splits as $\Pi = \Pi_0 + \Pi_{1|0}$ with $\Pi_0\Pi_{1|0} = 0$.

Theorem 6.17

In this notation,

(a) $X_0\hat\gamma_0 = \Pi_0\mathbf{Y}$, $X_{1|0}\hat\beta_1 = \Pi_{1|0}\mathbf{Y}$, i.e. $\hat\gamma_0 = (X_0'X_0)^{-1}X_0'\mathbf{Y}$ and $\hat\beta_1 = (X_{1|0}'X_{1|0})^{-1}X_{1|0}'\mathbf{Y}$, and $\hat\beta_0 = \hat\gamma_0 - (X_0'X_0)^{-1}X_0'X_1\hat\beta_1$;

(b) $\operatorname{Cov}(\hat\gamma_0, \hat\beta_1) = 0$, $\operatorname{Var}\hat\gamma_0 = \sigma^2(X_0'X_0)^{-1}$ and $\operatorname{Var}\hat\beta_1 = \sigma^2(X_{1|0}'X_{1|0})^{-1}$.

Proof. (a) In the parametrization (γ_0, β_1) the design matrix is $(X_0, X_{1|0})$ with $X_0'X_{1|0} = 0$, so its Gram matrix is block diagonal and $(X'X)^{-1}X'\mathbf{Y}$ splits into the two stated pieces; the relation for $\hat\beta_0$ inverts the definition of γ_0.

(b) $\operatorname{Cov}(\hat\gamma_0, \hat\beta_1) = \sigma^2(X_0'X_0)^{-1}X_0'X_{1|0}(X_{1|0}'X_{1|0})^{-1} = 0$, and the variances follow from Theorem 6.16(a) for each block. ∎

Example 6.17 (Centring the explanatory variables)

With an intercept, $EY_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} = \gamma_0 + \sum_j \beta_j(x_{ij} - \bar x_{\cdot j})$, where $\gamma_0 = \beta_0 + \sum_j \beta_j \bar x_{\cdot j}$. Here $X_0 = \mathbf{1}$ and $X_{1|0}$ has entries $x_{ij} - \bar x_{\cdot j}$. With $S_{jk} = \sum_i (x_{ij} - \bar x_{\cdot j})(x_{ik} - \bar x_{\cdot k})$ and $S_{jY} = \sum_i (x_{ij} - \bar x_{\cdot j})(Y_i - \bar Y)$,

$$\hat\gamma_0 = \bar Y, \qquad \begin{pmatrix} S_{11} & \cdots & S_{1p} \\ \vdots & & \vdots \\ S_{p1} & \cdots & S_{pp} \end{pmatrix} \begin{pmatrix} \hat\beta_1 \\ \vdots \\ \hat\beta_p \end{pmatrix} = \begin{pmatrix} S_{1Y} \\ \vdots \\ S_{pY} \end{pmatrix}.$$

For $p = 1$ this is $\hat{\beta}_1 = S_{1Y}/S_{11}$ and $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$, with $\operatorname{Var} \hat{\beta}_1 = \sigma^2/S_{11}$ and $\operatorname{Cov}(\bar{Y}, \hat{\beta}_1) = 0$.